Monitoring Measurable Residual Disease in ALL and AML

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57.1 Monitoring MRD in ALL

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57.1.1 Introduction

In ALL evaluation of molecular treatment response, assessment of minimal residual disease, nowadays named measurable residual disease (MRD), is a substantial independent predictor of outcome, as proven by randomized studies (Conter et al. 2010; Gökbuget et al. 2012; Bassan and Spinelli 2015). Consequently, MRD is implemented in virtually all clinical protocols in order to supplement or to redefine multifactorial risk stratification with optional customized treatment intensity. The detection of leukemic cells below the limit of classical cytomorphology is feasible by either disease-specific alterations of the immune phenotype or unique genetic features. Several competing and complementing MRD methods have been developed with preference application according to clinical protocols (Van der Velden et al. 2007; van Dongen et al. 2015).

57.1.2 MRD Assessment by IG/TCR Real-Time PCR

The discontinuous immune receptor genes provide the immune repertoire by somatic recombination of variable (V), diversification (D), and junction (J) elements, thus forming hypervariable CDR3 (complementarity-determining region 3) regions during lymphocyte maturation. Such rearrangements can serve as clonal index of leukemia blasts originating from lymphoid precursor stages. Additionally, due to a relaxed regulatory control, leukemia blasts can harbor incomplete rearrangements and cross-lineage rearrangements and tend to accumulate simultaneously multiple rearrangements. Quantitative real-time PCR using junction complementary allele-specific oligonucleotides (ASO) frequently reaches a detection limit of 1E-05 with a quantitative range of 1E-04, is applicable to vast majority of cases, and has a high degree of standardization (Van der Velden et al. 2007).

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57.1.3 MRD Assessment by IG/TCR Digital PCR

One significant drawback of qPCR is its relatively high level of inaccuracy, susceptibility to inhibiting substances, and reliance on reference samples for MRD quantification. A novel technique, known as digital PCR and considered the third generation of PCR (Vogelstein and Kinzler 1999; Starza et al. 2022), has been developed. Although the digital PCR approach shares similarities with qPCR in terms of targets and primers, it can overcome these limitations. However, achieving precision in digital PCR relies on maintaining a well-balanced sample load in the reaction compartments (Huggett et al. 2013). The clinical usefulness of this method is still being verified (Della Starza et al. 2021).

57.1.4 MRD Assessment by Fusion Gene Transcript

Most frequent recurrent reciprocal translocations are in ALL t(9;22)(q34;q11) (*BCR-ABL1*), t(12;21)(p13;q23) (*ETV6-RUNX1*), and t(4;11)(q21;q22) (*MLL-AFF1*) with age stage-associated preponderance in adults, childhood, and infant ALL, respectively. Derived chimeric fusion transcripts are validated marker for MRD detection by real-time PCR with an achievable detection limit of 1E-06. The methodology has been standardized by the European Against Cancer (EAC) program (Gabert et al. 2003).

57.1.5 NGS (Next-Generation Sequencing)

High-throughput sequencing (HTS) of immune receptor genes by next-generation sequencing (NGS) is a novel option for MRD. This methodology provides comprehensive qualitative and quantitative information regarding clonal consistence of the diagnostic sample and shares one protocol for index determination and MRD assessment without the need of individual reagents. Potential subleukemic and new emerging leukemic clones also are covered. PCR steps during library construction can introduce bias effecting results internal controls and normalization calculations are necessary the generated data volume is high and data interpretation demand biostatistics expertise. Due to high sample capacity, NGS favors a centralized concept, and service is available to commercial providers by academic centers (Kotrova et al. 2015).

57.1.6 Flow Cytometry

MRD by multicomponent flow cytometry (MFC) distinguishes leukemia-associated immune phenotypes (LAIP) and regular cells. LAIP consists of cell lineage maturation stage-specific (backbone) markers in combination with illegitimate markers. The standard four- to six-color approaches have been developed simultaneously by several centers. Therefore, the applied marker panels depend on study protocol. The consistently achieved detection limit is 1E-04. Recently, increase of specificity and sensitivity was enabled by high-throughput procedures demanding eight- or ten-color equipment. Here, the options for targeted and visualized antigens allow simultaneous visualization of all developmental lymphocyte stages serving as background to distinguish leukemic cells. The EuroFlow Consortium validated available antibody panels and controls which can be applied in a standardized way, including automated gating with supportive software, data storage and comparison, accurate quantitative result, and option for IVD development. Similar to the NGS approach, the generated data volume is high, and data interpretation demands biostatistics expertise; nevertheless, the concept allows decentralized data acquisition (Pedreira et al. 2013).

57.1.7 Limitations of MRD Assessment

The determined level of MRD always is a result of complex interrelation of baseline characteristics of tumor and patient, time point of MRD

evaluation, therapeutic agents, course of clearance, and degree of therapy resistance. Several measurements therefore are mandatory. Adverse circumstances for MRD assessment are clonal selection and clonal evolution, since the associated index might be missed. Potentially impacted are leukemia with initial oligoclonality as observed in approximately 15% of B-ALL, and up to 1000 subclones have been reported (Wu et al. 2016). Phenotypic plasticity under treatment and massive lymphocyte regeneration can cause false negativity or positivity, a solvable problem by applying mentioned high-throughput methodologies. Achievable detection limit is correlated with cell count of sample, and aplastic samples are challenging. Finally, all methodologies use different sample preparations, and analyses refer to different units, a circumstance which interferes result comparison.

57.1.8 MRD in the Setting of HCT

As all adult patients with ALL who relapse after initial chemotherapy have an absolute indication for allo-HCT, pediatric patients are stratified into different treatment groups. Main prognostic determinants in these patients are the blast immune phenotype, time to relapse, and site of relapse. High-risk patients who experienced early isolated BM relapse, early relapse involving BM, and any BM relapse of T-lineage ALL have clear indications for HCT. Intermediate-risk patients experienced early or late combined BM relapse and a late isolated BM relapse of a B-cell precursor (BCP).

ALL and very early and early isolated extramedullary relapse of either BCP-ALL or T-ALL have indication for HCT if post-induction MRD exceeds a threshold of 1E-03 (Eckert et al. 2013).

During the past decades, it could be clearly shown by several studies that the level of MRD immediately prior to transplant does have a clear prognostic impact on post-HCT outcome (Knechtli et al. 1998). Retrospective studies in children with relapsed ALL revealed an important cutoff for post-HCT outcome. Patients who received transplantation with an MRD load of ≥10 to 4 leukemic cells had a by far inferior prognosis than patients with lower MRD loads before transplant (Bader et al. 2009). Based on these findings, several studies are now underway investigating strategies to improve outcome in these ultrahigh-risk patients. Adaption of transplant approaches might allow successful transplantation (Leung et al. 2012).

Spinelli et al. showed that almost half of the patients with high levels of MRD before transplantation achieved molecular remission by day +100 (Spinelli et al. 2007). This finding indicates that MRD detection posttransplant provides additional value to the MRD assessment prior to transplantation. It could be demonstrated in prospective clinical studies that the close monitoring of MRD by different approaches allows the prediction of relapse and may therefore form the basis of different intervention strategies making use of leukemia-specific targeted therapy (Bader et al. 2015; Balduzzi et al. 2014). Future perspectives will focus on MRD-guided intervention to prevent overt relapse (Rettinger et al. 2017).

57.2 MRD in AML

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57.2.1 Introduction

Defining residual disease below the level of 5% leukemic cells has changed the landscape of risk classification (Ossenkoppele 2013). The measurable residual disease (MRD) approach establishes the presence of leukemia cells down to levels of $1{:}10^3$ to $1{:}10^6$ white blood cells, compared to 1:20 for morphology. The ELN 2022 recommendation response criteria includes apart from CR without measurable residual disease (CR_{MRD-}) now also CRiMRD− and CRhMRD− and is defined as CR,Cri/CRh with negativity for a genetic marker by RT-qPCR or CR with negativity by multicolor flow cytometry (MFC) (Döhner et al. 2022).

The reasons to apply MRD assessment in AML are (1) to provide a quantitative methodology to establish a deeper remission status; (2) to

better predict outcome and guide post-remission treatment; (3) to identify early relapse as a robust posttransplant surveillance, in order to enable early intervention; and (4) in the future to serve as a surrogate endpoint for survival to accelerate drug testing and approval (Ossenkoppele and Schuurhuis 2016).

The recent ELN MRD consensus document includes leads for standardized multiparameter flow cytometry-based MRD (MFC-MRD) and molecular MRD, MRD thresholds, and guidelines for clinical implications (Heuser et al. 2021).

57.2.2 Methods for MRD Detection

57.2.2.1 MRD Detection by PCR

Real-time quantitative PCR (RT-qPCR) allows MRD detection in cases with chimeric fusion genes generated by balanced chromosomal rearrangements (Grimwade and Freeman 2014). Other genetic alterations can also be used for MRD detection including mutated NPM1, RUNX1-RUNX1T1, CBFBMYH11, PML-RARA, KMT2A-MLLT3, DEK-NUP214, BCR-ABL, and WT1. Apart from t(15;17) and *RUNX1–RUNX1T1* and *CBFB–MYH11*, currently, *NPM1* is the best-validated molecular marker for MRD assessment. PCR assessment of MRD is in about 40–60% of patients in principle possible. The methodology has been standardized for several molecular markers for clinical implementation in the Europe Against Cancer (EAC) program (Gabert et al. 2003).

57.2.2.2 Immune MRD by Multicolor Flow Cytometry

The basic principle is to integrate diagnostic leukemia-associated immune phenotypes (LAIP). and different-from-normal (DfN) aberrant immunophenotype approaches to enable tracking of diagnostic and emergent leukemic clones (Heuser et al. 2021). These LAIPs consist of normally occurring markers, present in aberrant combinations in AML but in very low frequencies in normal and regenerating BM. The background levels of LAIP in normal and regenerating BM levels, in particular, although low, prevent specific detection of aberrancies with sensitivities higher than 1:10,000.

If no diagnosis sample is present, one can make use of "different-from-normal" approach which uses a standard fixed antibody panel to recognize leukemic cells based on their difference with normal hematopoietic cells (Loken et al. 2012).

Currently, immune MRD aberrancies can be detected in over 90% of AML cases at diagnosis.

57.2.2.3 MRD Detection by NGS

NGS-based molecular MRD assessment targeted NGS-based MRD testing using specific mutations identified at diagnosis vs agnostic panel approaches are now mostly exploratively applied. Diagnostic AML samples are generally screened for mutations using a multigene panel (Jongen-Lavrencic et al. 2018).

Prognostic impact has been shown for selected mutations present at diagnosis and/or in complete remission (CR) samples (Ghannam et al. 2020; Dillon et al. 2023). Germline mutations (ANKRD26, CEBPA, DDX41, ETV6, GATA2, RUNX1, andTP53) are noninformative as NGS-MRD markers (Godley 2021).

DMT3A, TET2, and ASXL1 (DTA) mutations can be found in age-related clonal hematopoiesis and like germline mutations should not be used for MRD analysis as these mutations often persist during remission and do usually not represent the leukemic clone (Shlush 2018; Hasserjian et al. 2020).

Recently, three studies showed that FLT3-ITD is a highly prognostic biomarker in AML patient when measured after induction chemotherapy (Grob et al. 2023; Loo et al. 2022; Dillon et al. 2020). By a bioinformatic approach, FLT3-ITDs can be reliably detected with NGS (Blätte et al. 2019). These studies uniformly show that FLT3-ITD MRD $\geq$ 0.01% is clearly associated with outcome.

57.2.3 MRD in Clinical Studies

Despite a multitude of prognostic factors at diagnosis, the outcome of patients is still highly

variable and not individually predictable. On-treatment parameters in combination with prognostic factors present at diagnosis may be more useful.

The prognostic value of MRD in remission has been shown in patients treated with both intensive and more recently less-intensive treatment modalities (Terwijn et al. 2013a; Maiti et al. 2021; Freeman et al. 2013; Pratz et al. 2021). A recent systematic meta-analysis of 81 publications has convincingly shown the prognostic value of MRD for relapse and overall survival (Short et al. 2020). However, MRD is far from perfect, since relapses still occur in MRD-negative patients. Thus, a negative MRD test result may not indicate complete disease eradication but refers to disease below the MRD test threshold in the tested sample. Conversely, not all patients who are MRD-positive will relapse. Of note, Mol-MRD may remain detectable at low levels (CRMRD-LL) without prognostic significance and, therefore, are called negative operationally if the MRD values are below the threshold linked to prognosis (Dillon et al. 2020; Heuser et al. 2021). For instance, in CBF-AML and *NPM1* mutant AML, the transcripts may show persistent low-level expression after treatment, but this is not prognostic of relapse (Freeman et al. 2018).

Unfortunately, surrogacy for survival has not been proven yet (Hourigan et al. 2017; Ossenkoppele and Schuurhuis 2016; Walter et al. 2021).

57.2.4 MRD in Relation to Transplant

Evidence is accumulating that the presence of MRD assessed by multicolor flow cytometry immediately prior to allogeneic HCT is a strong independent predictor of posttransplant outcomes in AML (Buckley et al. 2017; Walter et al. 2015). Araki et al. showed that in 359 adults, the 3-year relapse rate was 67% in MRD-positive patients, compared to 22% in MRD-negative patients, resulting in OS of 26% vs. 73%, respectively (Araki et al. 2016). This applies for the myeloablative as well as for the non-myeloablative transplant setting. The same was found in a large EBMT study (Gilleece et al. 2018).

Also, molecular MRD as measured by RT-PCR in NPM1-mutated AML has a significant impact on outcome after allo-HCT (Balsat et al. 2017).

Pretransplant MRD is prognostically also useful in CR2 (Gilleece et al. 2020). Importantly, it was demonstrated that the intensity of the conditioning regimen is dependent on MRD status. Myeloablative conditioning was only advantageous in the MRD+ setting (MRD+, RIC 34% vs MAC 59% 3 years OS; MRD−, RIC 61% vs MAC 60% 3 years OS) (Hourigan et al. 2020). We recently showed that MRD-guided therapy in intermediate-risk AML patients is a valuable strategy in reducing the number of allogeneic transplants without negatively affecting survival (Tettero et al. 2023).

Only sparse data is available for the posttransplant situation. Most available studies showed that the presence of posttransplant MRD had an adverse prognostic impact (Klyuchnikov et al. 2022; Zhou et al. 2016). In one study, this was irrespective of pretransplant MRD in patients with AML (Loke et al. 2023). Many trial groups certainly in Europe decide which post-remission treatment should be given based on MRD. Allo-HCT in the favorable group is usually not applied in the MRD-negative setting. For the intermediate group MRD− AML patients do not get an alloHCT in many centers and is only applied in CR2. The adverse-risk patient receives an allo-HCT independent of MRD status. The value should still be prospectively proven.

57.2.5 Clinical Intervention Studies Posttransplant

There is a current much interest on intervention studies in the posttransplant setting based on MRD assessment. Preemptive therapy with azacitidine can prevent or substantially delay hematological relapse in MRD-positive patients with MDS or AML who are at high risk of relapse (Platzbecker et al. 2018). A number of groups summarized by Biederstadt et al. have explored administering DLI preemptively on detection of measurable residual disease (MRD) or mixed chimerism. Evidence for the effectiveness of this

strategy, although encouraging, comes from only a few, mostly single-center (Biederstädt and Rezvani 2023). Also, application of targeted therapy after transplant (e.g., FLT3 inhibitors) is currently under investigation .

It is clear that novel treatment strategies before, during, and after transplant are urgently needed to improve outcomes in AML. Thereby, depth of response prior to transplant, as measured by level of MRD, has emerged as one of the most important predictors of transplant outcome. Randomized trials are warranted to determine if MRD-guided preemptive therapy is associated with improved outcome.

Most importantly no clinical trial including transplantation trials should be performed without including MRD assessment.

57.2.6 Future Developments

New technologies are emerging to assess MRD. Standardization and harmonization are important and are currently further explored by the ELN MRD WP (Tettero et al. 2021). Quantifying leukemic stem cells seems a promising approach (Ngai et al. 2023; Terwijn et al. 2014; Zeijlemaker et al. 2016). MRD-guided studies like post-remission decisions, MRD conversion pretransplant, MRD-directed intensification of conditioning regimens, and posttransplant intervention should be encouraged.

Key Points

- MRD is now included in the definition of CR.
- MRD positivity is an independent predictor of relapse after chemotherapy in AML patients and a negative predictor for ALL patients.
- Pretransplant MRD positivity is highly indicative for relapse.
- MRD assessment should be implemented in every clinical trial.
- Prospective intervention studies guided by MRD are being performed.

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Further Readings

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